

CSA0619-Design and Analysis of Algorithm

CO3 ASSESSMENT TOOL2 Case Study Based Assessment

28. Online Book Search System using Binary Search:

main.py	Output
<pre>1 # Binary Search - Online Book Search System 2 3 books = input("Enter sorted book names: ").split() 4 target = input("Enter the book to search: ") 5 6 low = 0 7 high = len(books) - 1 8 found = False 9 10 while low <= high: 11 mid = (low + high) // 2 12 13 if books[mid] == target: 14 print("Book found at position", mid) 15 found = True 16 break 17 elif books[mid] < target: 18 low = mid + 1 19 else: 20 high = mid - 1 21 22 if not found: 23 print("Book not found")</pre>	<pre>Enter sorted book names: python java sql Enter the book to search: sql Book found at position 2 === Code Execution Successful ===</pre>

Analysis:

Binary Search compares the target book with the middle element. If the target is smaller, it searches the left half; otherwise, it searches the right half. This process continues until the book is found or the search space becomes empty.

Report Analysis:

Case	Time Complexity
Best Case	$O(1)$
Average Case	$O(\log n)$
Worst Case	$O(\log n)$

Space Complexity:

$O(1)$ (Iterative Binary Search)

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